Graphic Era HILL UNIVERSITY

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University under section 2(f) of UGC Act 1956

End Term (Even) Semester Regular/Back Examination May 2026

Roll no.......2418159.......................

Name of the Program and semester: B.Tech. CSE. AI-DS. CS/4th
Name of the Course: Programming in Java
Course Code: TCS-408
Time: 3 hours   Maximum Marks: 100

Note:
- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Describe the key features of Java in detail. Also, support at least four features with suitable programs to demonstrate their implementation. (CO1)

b. Discuss Jagged arrays with proper line of codes. Write a Java program to find all the pairs of two numbers whose sum equals a given sum value. (CO1)

Input: Array: 2 4 7 5 9   Sum = 9

Output: 2+7=9 and 4 + 5 = 9

c. Explain the internal architecture of the Java Virtual Machine (JVM) with a neat and clean diagram. Also explain the role and functionality of each component of it. (CO1)

Q2. (2X10=20 Marks)

a. Discuss inheritance in Java and describe its types using proper codes. Also, discuss the limitation of multiple inheritance in Java and how it is resolved. (CO2)

b. Explain the concept of packages in Java and explain how they help in organizing large applications. Describe the process of creating and using packages with suitable examples. Also, differentiate between import and static import statements with appropriate codes. (CO2)

c. Explain and compare abstract classes and interfaces in Java based on their features, usage, and differences with suitable examples. Also write a Java program for the following scenario. (CO2)
A banking system needs to calculate simple interest using standard formula; write a Java program using an interface that declares method for interest calculation. Implement this interface in different class, apply the appropriate formula to compute and display interest.
Simple Interest (SI)= (P*R*T)/100   Where:
P= Principal amount
R= Rate of interest (per annum)
T= Time (in years)

Q3. (2X10=20 Marks)

a. Discuss the concept of exception handling in Java and its importance. Describe the types of exceptions, and different keywords given to handle it. Also write a program for the following scenario.
In a banking system users perform operations such as withdrawal and deposit. Create a custom exception class, *MinimumBalanceException* to handle cases where a withdrawal causes the balance to fall below a specified minimum limit. (CO3)

b. Explain the difference between byte streams and character streams. Also write a program that counts all the vowels present in a given file. (CO3)

c. Explain the concept of multithreading in Java and describe the life cycle of a thread with a neat and clean diagram. Also write a program to set and get the priorities of a thread. (CO3)

Q4. (2X10=20 Marks)

a. Discuss the Collection and Generic Framework in Java. Write a Java code to create ArrayList of integers with initial values 18,23,48,67,88 Then insert 98, 12 at the end and 35 at index 3. Also, write the code to traverse the ArrayList using iterator. (CO4)

b. A client wants a GUI for the Employee's information. Write a Java Swing program using labels, text fields, buttons, radio buttons, text-areas, checkboxes, etc. to provide the solution (Only GUI). (CO4)

c. Discuss the different JavaFX layouts and write the lines of code used to set them. Also, write a JavaFX application that displays a TextField, a Label, and a Button. (CO4)

Q5. (2X10=20 Marks)

a. Explain the significance of JDBC drivers in Java and describe their different types with a neat and clean diagram, along with their advantages and disadvantages. (CO5)

b. Create a GUI application using Swing in Java for a simple Addition Calculator. The interface should include JLabel components for "Number 1", "Number 2", and "Result". JTextField components to accept numeric input and to show result, and a JButton labeled "Add".

Implement event handling such that when the button is clicked, the program reads the values entered in the first two text fields, calculates their sum, and displays the result in another JTextField on the form. (CO5)

c. Develop a simple Java application using JDBC to connect to a database. Consider a table named Student (already exist) with fields such as id, name, and marks. (CO5)

Write a program to:

(i) Establish a connection to the database

(ii) Insert at least two student records using JDBC

(iii) Retrieve and display all records from the table Student